1982ApJ...252..697H

THE ASTROPHYSICAL JOURNAL, 252:697–714, 1982 January 15
© 1982. The American Astronomical Society. All rights reserved. Printed in U.S.A.

TIME SERIES INFRARED SPECTROSCOPY OF THE MIRA VARIABLE χ CYGNI

KENNETH H. HINKLE, DONALD N. B. HALL, AND STEPHEN T. RIDGWAY
Kitt Peak National Observatory[1]
Received 1980 July 11; accepted 1981 July 23

ABSTRACT

A time series of 32 high resolution infrared spectra spanning more than three cycles of the S-type Mira variable χ Cygni is analyzed. Most spectra are of the 1.6–2.5 μm region; five are of a narrow region at 4.6 μm. Using information derived from the $\Delta v = 1$, 2, and 3 vibration-rotation bands of CO, we show that χ Cygni is characterized by a regularly pulsating photosphere and complex, circumstellar envelope.

Velocities, excitation temperatures, and column densities derived from the CO $\Delta v = 3$ bands around 1.6 μm show that the photometric variability is associated with an outwardly propagating wave, probably driven by a general stellar oscillation, which traverses the photosphere during the premaximum and maximum phases. A shock front forms between gas infalling from the previous cycle and material already accelerated by the emerging wave, and, at this interface, the gas is heated from $\sim$2000 to $\geq$4000 K, molecules are dissociated, and hydrogen is probably ionized. Two months after maximum light, the shock has passed through the photosphere, and, from phase 0.1 to 0.8, the material is uniformly inwardly accelerated at 0.098 cm s^{-2} while cooling monotonically.

The structure of the circumstellar envelope appears to be dominated by a stationary layer which is inferred from its 800 K temperature to be at a distance of the order of 10 stellar radii from the photosphere. This layer in χ Cygni built up rapidly late in 1975 and steadily diminished over the next three cycles. During this interval, it appears to have provided a reservoir both for material outwardly accelerated (presumably by radiation pressure on condensing dust) and for material which may be draining back down to the photosphere. At optical wavelengths, the resonance lines arise in the first of these regimes, and others arise in the second. The characteristic velocities of SiO maser features imply they arise on the outer boundary of the "800 K" layer.

Subject headings: infrared: spectra — stars: circumstellar shells — stars: individual — stars: long-period variables — stars: pulsation

I. INTRODUCTION

Blue-violet spectra of long period variable stars (LPVs) extending over many cycles show periodic changes in the intensities of both the absorption and emission lines that are correlated with the pronounced photometric variability (Merrill 1960). An outstanding problem in understanding the physical nature of long period variability is that the velocity behavior of the emission lines occurring near the time of maximum light is consistent with a pulsating photosphere characterized by shocks, while the absorption lines (Joy 1954) do not exhibit significant velocity variations with phase. The proposal by Reid and Dickinson (1976) that the centroids of broad thermal emission lines seen at millimeter wavelengths represent the true stellar velocity implies the blue-violet absorption lines are systematically red-shifted[2] with respect to the stellar velocity (Wallerstein 1979). The failure of the blue-violet absorption lines to reflect the kinematics of a propagating disturbance is particularly disconcerting because models of the variability invoking a general oscillation of the stellar interior are consistent with the emission line behavior around maximum light and predict easily observable velocity changes in absorption lines (Wood 1979). Observations of maser lines have failed to shed light on the nature of the variability, revealing instead a complex circumstellar envelope.

The availability of Fourier transform spectrometers at large telescopes has made it possible to obtain high resolution ($\lambda/\Delta\lambda \approx 50{,}000$) spectra in the 1.5 to 5 μm region of the infrared. This spectral region is particularly suitable for studying Mira variability since it in-

[1]Operated by the Association of Universities for Research in Astronomy, Inc., under contract with the National Science Foundation.

[2]In this paper, the convention that "red" and "blue" shift refer to shifts to lower and higher frequencies is extended to lines in the infrared.

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1982ApJ...252..697H

cludes both the 1.6 μm window where H^- opacity is at a minimum and the $\Delta v = 1$, 2, and 3 vibration-rotation bands of CO around 4.6, 2.3, and 1.6 μm. These bands are ideal probes of the atmospheric structure because their lines span a wide range of excitation energies and strengths and there exist excellent laboratory data on frequencies and absorption coefficients. The density of lines is not nearly as high as in the visible, so that both unblended lines and regions of clean continuum are present. Miras are among the brightest objects in the sky in this region of the infrared and, since they vary by less than 1 mag, can be observed at any phase. Bright infrared objects may be observed equally well during day or night, so continuous phase coverage is possible.

An analysis of eight 1.5–2.5 μm spectra of the M8 Mira R Leonis demonstrated that the photospheric lines at these wavelengths do exhibit a velocity variation consistent with a shock wave propagating out through the photosphere around maximum light (Hinkle 1978). In the present paper, we present the analysis of a more comprehensive set of infrared spectra of the S-type Mira, χ Cygni. Over a 3 yr interval, beginning early in 1976, 27 1.5–2.5 μm spectra of χ Cygni were obtained at 1 or 2 month intervals. These are supplemented by five observations of the CO fundamental at 4.6 μm. Details of the observations are set out in § II.

The spectra provide continuous coverage over nearly three cycles, revealing a complex process with up to four distinct velocity components being distinguishable at one time in the CO lines. The 1.6 μm $\Delta v = 3$ lines are formed in the photosphere and mimic the R Leo scenario of an outwardly propagating shock wave around maximum light; their analysis is described in § III. The 2.3 μm $\Delta v = 2$ lines exhibit the same behavior and, in addition, reveal both a stationary component at a temperature of 800 K and also a hotter, infalling component; these results are detailed in § IV. The 4.6 μm $\Delta v = 1$ lines, which are excellent probes of distant circumstellar material, are treated in § V. In § VI, results from these diverse spectral regions are consolidated into an overall picture of the atmosphere and related to the comprehensive observations of χ Cygni at visible (Merrill 1947, 1953; Maehara 1971) and millimeter (Snyder and Buhl 1975) wavelengths. Key results are summarized in § VII.

II. SPECTRA

The 32 spectra used in the present analysis were obtained with the Mayall 4 m telescope between 1976 January and 1979 February. Early spectra were observed with a prototype Fourier transform spectrometer (FTS) similar to the instrument described by Ridgway and Capps (1974). In 1978 September the prototype was replaced by a more versatile and efficient spectrometer (Hall *et al.* 1979) which has since been used for all observations.

Both FTSs utilize rapid continuous scanning to record an interferogram at 0.07 cm^{-1} resolution in 1 or 2 minutes. Sampling by an analog to digital converter is referenced to a stabilized control laser, so frequencies in the interferogram are recorded as multiples of this laser's frequency. Successive scans are co-added and stored on disk until an adequate signal-to-noise ratio is obtained.

The intrinsically high photometric accuracy of the rapid scanning mode (where all points in a scan are sampled at essentially the same air mass, terrestrial velocity correction, etc.) is enhanced by a number of instrumental and observational techniques. In particular, the full symmetry of the Michelson interferometer is utilized, i.e., both inputs and both outputs. Use of two inputs allows first-order cancellation of spectra common to them, such as thermal emission or scattered sunlight. Switching of the source from one aperture to the other between scans with an accompanying change in the sign of the co-adding of the interferogram further improves cancellation. The signals from matched detectors at the two outputs are both summed (to remove the interferogram leaving the instantaneous stellar flux) and differenced (to yield an interferogram with double the amplitude and a much smaller DC offset). The difference signal, electrically filtered to the appropriate bandpass (400–1000 Hz), is then ratioed to the sum signal which has been electrically low pass filtered (DC-200 Hz) so as to eliminate amplitude modulation of the interferogram due to scintillation, guiding errors, etc. The sum signal level is also tested, and points are not co-added if the sum drops below a preselected fraction of its peak value.

As a result, the photometric accuracy and frequency calibration of the spectra are limited by white photon noise rather than instrumental characteristics. The frequencies of all points in a spectrum are expressed as multiples of the reference laser frequency and, after compensation for the dispersion of air, require only a single correction for slight collimation differences between the reference laser and signal beams.

The spectra obtained with the prototype FTS have not been convolved with an apodizing function since the interferometer yielded some instrumental apodization associated with systematic variation in reimaging onto the beamsplitter with path difference. This instrumental apodization is not indicated in Table 1, and the actual instrumental FWHM is $\sim 10\%$ greater than the theoretical values listed. The instrumental profile as indicated by sharp terrestrial lines remained stable over the life of the prototype instrument. Its replacement yields theoretical sinc instrumental profiles, and these spectra have been apodized by function I2 of Norton and Beer (1976) to reduce the instrumental profile sidelobes on the spectral lines.

Relevant parameters for each spectrum are summarized in Table 1. The spectra listed in Table 1A cover

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1982ApJ...252..697H

TABLE 1A

TWO MICRON REGION SPECTRA OF χ CYGNI

Date	JD (2,440,000+)	Visual Phase[a]	S/N[b]	Resolution[b] (theoretical FWHM)	Integration Time (minutes)
1976 Jan 12	2790.3	0.56	53	0.07	16
1976 Jan 14	2792.3	0.57	54	0.07	15
1976 Mar 5	2843.2	0.69	67	0.06	17
1976 Apr 7.....	2876.1	0.77	55	0.06	30
1976 Apr 12	2881.1	0.79	72	0.06	32
1976 May 11 ...	2910.1	0.86	77	0.08	11
1976 Jun 7.....	2937.1	0.90	64	0.06	32
1976 Jul 26	2985.9	0.04	28	0.06	37
1976 Aug 18 ...	3007.6	0.10	84	0.06	50
1976 Sep 29	3050.7	0.20	61	0.06	36
1976 Nov 6	3088.6	0.30	30	0.06	34
1976 Dec 5	3118.4	0.37	44	0.06	62
1977 Jan 11	3155.3	0.46	108	0.06	64
1977 Feb 8.....	3183.3	0.52	~40[c]	0.06	11
1977 Feb 19	3194.2	0.55	51	0.06	31
1977 Mar 28 ...	3231.2	0.64	45	0.06	35
1977 May 30 ...	3294.7	0.79	86	0.06	30
1977 Jun 24	3319.0	0.85	81	0.06	31
1977 Aug 26 ...	3381.7	0.01	84	0.06	60
1977 Oct 2.....	3418.6	0.10	151	0.06	60
1977 Oct 24	3441.5	0.15	96	0.06	32
1977 Nov 30 ...	3478.3	0.24	97	0.06	33
1978 May 27 ...	3656.0	0.67	120	0.06	64
1978 Oct 17	3798.6	0.02	280	0.08	68
1978 Dec 8	3851.5	0.15	91	0.06	81
1979 Feb 8.....	3913.2	0.30	260	0.07	79
1979 Feb 19	3924.2	0.33	72	0.06	62

[a]Phases computed from the nearest time of visual maximum light. JD of maxima used are (+2,440,000) 2555, 2968, 3379, 3791 (Mattei 1979).

[b]1976 Jan 12 to 1978 May 27 unapodized. 1978 Oct 17 to 1979 Feb 19 apodizing function I2 of Norton and Beer 1976.

[c]No reverse scan.

TABLE 1B

FIVE MICRON REGION SPECTRA OF χ CYGNI

Date	JD (2,440,000+)	Visual Phase	S/N[a]	Resolution[a] (theoretical FWHM)	Integration Time (minutes)	Bandpass (cm^{-1})
1977 Feb 9.....	3183.2	0.52	67	0.06	25	2100–2200
1977 Oct 2.....	3419.7	0.10	174	0.06	30	2100–2200
1978 Feb 17	3557.1	0.44	80	0.07	20	2100–2200
1978 May 22 ...	3651.1	0.66	75	0.06	17	2100–2200
1978 Oct 18	3799.7	0.02	70	0.06	33	2000–2200

[a]Unapodized.

the range 4000–6600 cm^{-1} and include $\Delta v = 2$ and $\Delta v = 3$ vibration-rotation bands of CO. They do not represent the highest quality data presently obtainable, but the typical signal-to-rms noise ratios of 70 and spectral resolution ≈ 5 km s^{-1} are well suited to the present analysis. The narrowest photospheric lines have a FWHM twice the instrumental resolution (§ III*b*). Five observations (Table 1B) are of the CO $\Delta v = 1$ region around 2140 cm^{-1} and were obtained through cooled, 100 or 200 cm^{-1} bandpass interference filters so as to reduce shot noise due to thermal background radiation. The velocity resolution in these longer wavelength spectra is ~ 10 km s^{-1}. The signal-to-noise figures given in Table 1 are the peak signal compared to the rms noise in the difference between pairs of spectra transformed from interferograms obtained with the interferometer scanned in opposite directions. The 4000–6600 cm^{-1} spectra were obtained through a 7″.5 aperture, and the 2140 cm^{-1} spectra through a 2″.8 aperture. The loss of signal as a result of light not entering the

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1982ApJ...252..697H

aperture under poor observing conditions is reflected in some of the signal-to-noise values in Table 1.

III. THE SECOND OVERTONE ($\Delta v = 3$) BANDS

In late-type stellar spectra, several hundred unblended lines are present in the CO second overtone, making this sequence ideal for both radial velocity and curve of growth analysis. In the M giants, including the Miras, the CO second overtone lines are of moderate strength. Our spectra show that the CO is considerably stronger in the S-type Miras than in the M types, probably because of decreased continuous opacity (Hall *et al.* 1982). In χ Cygni, the strongest second overtone CO lines have central depths of between 70% and 75% at most phases.

As the first step in the analysis, the profiles of a group of 30 normally unblended $\Delta v = 3$ CO lines were combined to produce an "average" line profile. The 30 lines from the 4–1, 5–2, 6–3, and 7–4 bands, with J-values ranging from 4 to 41, were selected from among the strongest second overtone lines. The composite profile resulting from the averaging indicates the velocity of the second overtone lines and the general shape of the line profile. The method described by Hinkle (1978) was used to set the continuum through the high points. Care was taken to ignore emission lines, and the continua seem consistent from phase to phase and cycle to cycle.

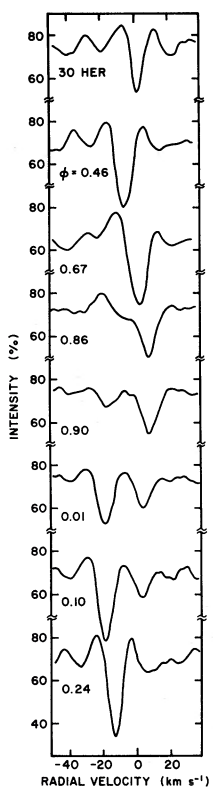

FIG. 1.—Average CO second overtone line profiles as a function of phase. The velocity scale is heliocentric. The intensity at the continuum is 100%. The M6 III low amplitude variable 30 Her is shown for a comparison profile.

Figure 1 presents a set of average profiles at various phases. Note that since the individual lines were initially selected as unblended, the edges of the average profile are higher than regions well away from the lines. This is enhanced by the damped sinc function instrumental profile. An average profile generated from the spectrum of a low amplitude, variable M giant is presented as an example for a more "normal" star.

Four basic features of the spectral lines can be seen on the plots of average profiles. (1) The spectrum is double lined for about one-fourth of the light cycle, from before phase 0.90 until about phase 0.15. During the double-line phases, the line to the red fades as the line to the blue strengthens. (2) The absorption lines move to the red with phase. (3) A blueshifted emission edge develops after phase 0.60 and emerges as a separate emission feature. This appears to invert into the blueshifted absorption line of the doubled absorption line phases. (4) The lines weaken during the emission line and double-line phases. In the following subsections, more detailed measurements of the individual spectral lines will be described that further elucidate the phase-dependent changes. Changes in radial velocity from one cycle to another are very small (§ VI), and all observations have been merged, using the visual phase, into a composite cycle for this section.

a) *Radial Velocities*

By measuring the velocities of individual lines rather than an average line profile, information on the variation of velocity with the depth of line formation is available. Following Hinkle (1978), the absorption line central depths were measured at the velocities indicated by the average profile, and the least blended lines selected on the basis of the central depths. The velocities of the least blended lines were determined using the deepest point in a line profile interpolated to 4 m s^{-1} point spacing. The small systematic correction to the velocities due to the collimation difference between the interferometer reference laser and stellar beams (generally $\sim$1 km s^{-1}) was calibrated from the measured positions of telluric lines in the spectrum. The resulting velocities are listed in Table 2 and plotted against time Figure 2. These velocities have been corrected to heliocentric rest; to convert to LSR, add +17.98 km s^{-1}. The uncertainties given in Table 2 are $\pm 1\ \sigma$.

A slight dependence of velocity on excitation potential for CO $\Delta v = 3$ lines with $E'' < 10{,}000$ cm^{-1} (1.24 eV) can be seen in the eight highest signal-to-noise spectra. Since the velocity of the weaker lines is desired, and the weaker lines are the lines of higher excitation, when a

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1982ApJ...252..697H

TABLE 2

VELOCITIES FROM TWO MICRON REGION SPECTRA

JD (2,440,000+)	$^{12}C^{16}O\,\Delta v=3$		$^{12}C^{16}O\,\Delta v=2$ 2–0 High Excitation		2–0 Low Excitation[a]	3–1 Moderate Excitation		
2790.3	−2.46±0.20		−1.3		−10.4	−11.2	−0.8:	
2792.3	−2.42±0.22		+0.5		−9.6	−8.9	−0.1:	
2843.2	+3.04±0.15		+5.8		−9.1	−8.4	+0.9:	
2876.1	+5.46±0.17		+7.0		−8.7	−7.7	−0.5	+7.2:
2881.1	+6.59±0.16		+6.9		−9.4	−7.2	+3.4:	+9.4:
2910.1	+7.97±0.18		+8.4		−8.7	−6.8	0.0:	+7.0:
2937.1	+8.15±0.13	−17.04±0.20	+8.6		−8.0	−8.3	−0.6:	+6.9:
2985.9	+3.93±0.31	−21.00±0.15	+6.3	−21.1	−9.3	−17.8	−3.5:	+3.5:
3007.6	+3.54±0.24	−18.10±0.09	+7.0	−18.8	−7.6	−13.1:	−3.0:	+3.4:
3050.7	−14.26±0.08		−16.2		−9.0	−16.7	−0.4:	+3.0:
3088.6	−11.61±0.18		−14.4		−7.1	−12.8	−1.2	
3118.4	−10.10±0.19		−11.4		−7.4	−13.7	+0.3	
3155.3	−4.23±0.11		−7.4		−6.4	−11.2	−0.3	
3183.3	−3.89±0.20		−1.8		−10.0	−8.8	+0.2	
3194.2	−2.61±0.18		−2.3		−9.6	−8.5	−0.5	
3231.2	+0.35±0.15		+0.6		−6.4	−6.7	+0.6	
3294.7	+5.78±0.18		+5.6		−6.6	−0.1:		
3319.0	+6.20±0.15		+7.1		−6.6	+1.8:		
3381.7	+3.81±0.22	−18.59±0.12	+4.1	−21.8	−7.3	−16.7	+3.0	
3418.6	+2.31±0.42	−15.40±0.04	−17.4		−7.3	−18.8	+3.1	
3441.5	−13.79±0.05		−15.4		−7.2	−18.4	+1.6	
3478.3	−12.85±0.04		−13.3		−9.1	−17.0	−0.5	
3656.0	+2.59±0.22		+2.6		−4.3	−1.8		
3798.6	+3.85±0.32	−17.50±0.08	+8.2	−19.5	−0.1	−19.2	+3.2	
3851.5	−15.26±0.08		−17.1		−0.7	−19.1	+1.7	
3913.2	−10.97±0.07		−14.0		−2.6	−17.4	−1.3	
3924.2	−10.27±0.08		−13.1		−16.4 −3.0	−16.2	−0.1	

[a] Deepest point of strongest component.

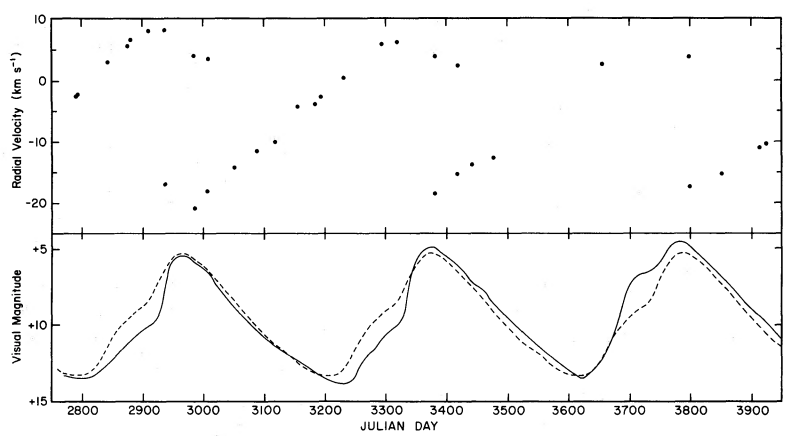

FIG. 2.—*Upper panel*: mean CO second overtone velocity as function of JD number (−2,440,000). Actual values with errors are listed in Table 2. *Lower panel*: visual light curve of χ Cygni as a function of JD number. *Dashed line*: average light curve (Campbell 1955). *Solid line*: actual light curve (Mattei 1979).

trend with excitation was noted, the velocity was determined from the lines with $E''>10{,}000$ cm^{-1}. The resulting change in the mean velocity is always less than 1 km s^{-1}. Seven of the eight spectra are of postmaximum phases (0.02 to 0.46), so little information on the phase dependence of the velocity-excitation potential relation is available. The top panel of Figure 3 illustrates a well-defined relation typical for phases 0.0 to 0.3. For phases 0.3 to 0.5, the relation becomes more extreme (Fig. 3, bottom). No trend of velocity with excitation potential can be discerned from the present premaximum spectra which all have lower signal-to-noise ratio.

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1982ApJ...252..697H

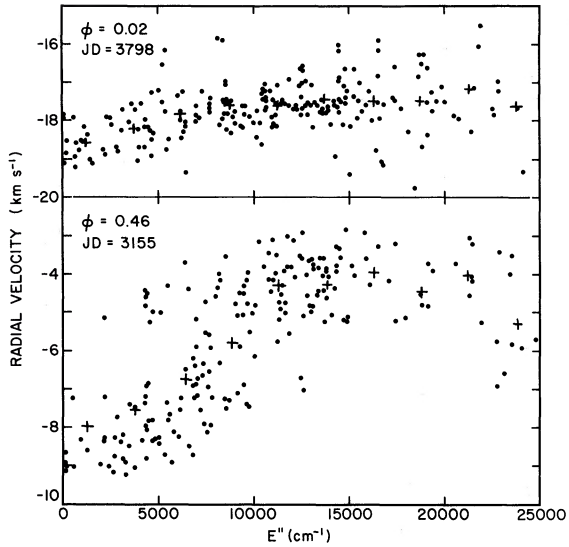

FIG. 3.—CO $\Delta v = 3$ radial velocities for individual lines as a function of lower-level excitation potential, E'', at two phases. *Pluses*: average velocity in 2500 cm^{-1} intervals. The large trend with excitation potential at phase 0.46 appears only near phase 0.4. Phase 0.02 presents a more typical case.

b) *Curves of Growth*

By a least-squares fit of a Gaussian to the deepest two-thirds of the line profile, the equivalent width was measured for each line selected as being unblended on the basis of central depth. The resulting equivalent widths were plotted as a function of central depth and the central depth calibrated in terms of equivalent width by fitting straight line segments through the lower (smaller equivalent width) envelope of the relation. This procedure considerably lessens the scatter in the curves of growth by removing the erroneously large equivalent widths resulting from those lines with blended wings but unblended cores.

Equivalent width versus central depth plots were produced for all 27 2 μm spectra. In a number of the lower signal-to-noise spectra, especially those with S/N less than 50, a well-defined relation between equivalent width and central depth is lacking. However, the 23 spectra obtained with the prototype FTS have nearly identical resolution. When the equivalent width–central depth plots were compared, it was clear that although the lines may broaden slightly during the minimum light to pre-maximum phases, one width-depth relation can be applied to all the prototype FTS spectra within the present uncertainties. The 1977 November 30 calibration was selected as being both the most representative and best defined and was used for the 23 prototype FTS spectra. The approximate FWHM of weak lines in this calibration, after correction for instrumental profile, is 0.19 cm^{-1}.

The four spectra taken with the new FTS are of both higher resolution and higher signal-to-noise than those of the prototype. For these spectra, individual central depth to equivalent width calibrations were used. From the three high signal-to-noise spectra of phases 0.02 (1978 October 17) and 0.30 (1979 February 8), the FWHM of the weak lines appears to be slightly decreasing from maximum to minimum light, ranging from 0.23 to 0.21 cm^{-1}. Both these and the 1977 November 30 spectrum indicate typical FWHM values near 10 km s^{-1} for the weak lines.

Having calibrated the central depths in terms of equivalent width, curves of growth were constructed following the procedure of Hinkle, Lambert, and Snell (1976). Discussions of curve of growth analyses for CO in cool stellar atmospheres can be found in that paper and in Dominy, Hinkle, and Lambert (1982). The excitation temperature was set by minimizing the difference between the high and low excitation lines on the curve of growth. In Figure 4, a series of curves of growth are displayed as a function of phase. It seems likely from the large equivalent widths and central depths of the CO $\Delta v = 3$ lines in χ Cygni that the strongest lines are well onto the flat portion of the curve of growth. Contrast between the boundary and photospheric temperatures almost certainly governs the depths of the strongest lines.

The excitation temperature is shown as a function of phase[3] in Figure 5 and the measured values are listed in Table 3. The excitation temperature as a function of phase is well defined, dropping nearly 2000 K during the course of the light cycle. The measurements can be represented by the expression $T_{exc} = 3600 \exp(-t/920)$, where t is in days past maximum visual light. Note that several cycles have been folded together in Figure 5 and are in excellent agreement. The uncertainties on the excitation temperatures indicated by the scatter in the curves of growth are typically ± 200 K. However, the consistency in the plot of temperature versus phase indicates that this is an overestimate. Systematic errors in T_{exc} due to assumptions made in the curve of growth analysis should be $\leq 10\%$ (Hinkle, Lambert, and Snell 1976).

The shifts between the CO curves of growth at different phases give the change in the column density; this has been plotted as a function of phase in Figure 6. Order of magnitude changes in the CO column density during the light cycle are indicated, with large column density changes near maximum light and a value constant to ± 0.15 in $\log NL$ from about phases 0.20 to

[3]The full sequence of events associated with the propagation of each disturbance through the photosphere spans much more than a single period of the optical variability. In order to avoid confusion, we have specified the phase of events which occur in the preceding or following cycles by subtracting or adding 1.0 to the normal phase convention.

© American Astronomical Society • Provided by the NASA Astrophysics Data System

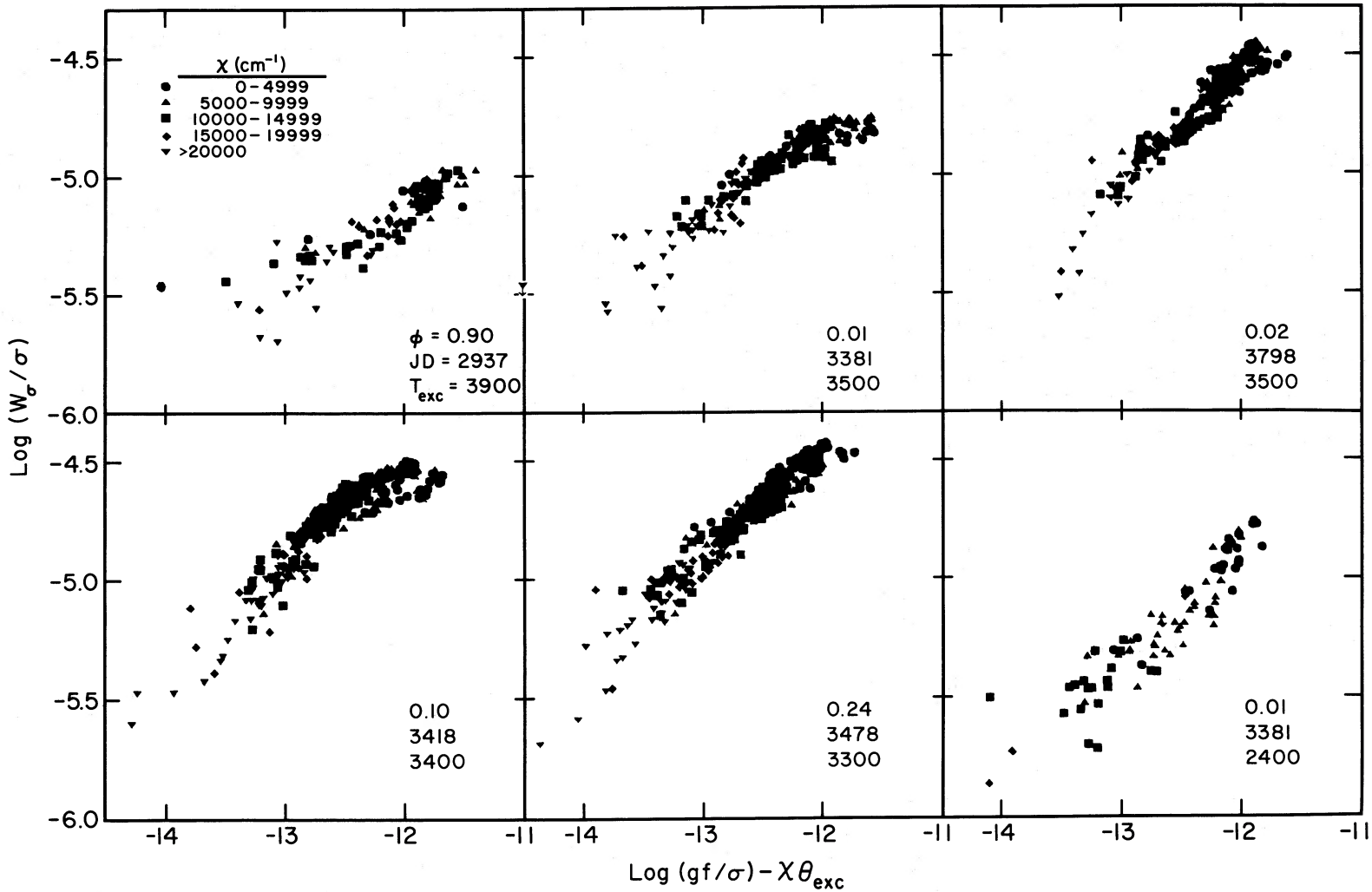

FIG. 4.—Curves of growth for the second overtone CO lines at six different phases. All the curves except for the lower right-hand corner are for the hot expanding gas. Note the large cyclic differences between the cycle represented by JD 3381, 3418, and 3478 and that represented by JD 3798. Symbols (*key*) indicate the excitation potential of the lower level of the transition.

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1982ApJ...252..697H

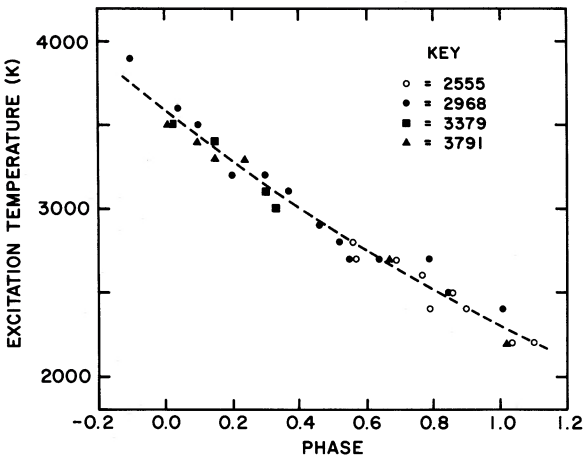

FIG. 5.—CO $\Delta v=3$ excitation temperatures as a function of visual phase. Dashed line is the fit described in the text. Symbols relate the measurement to light cycles starting on the indicated JD ($-2{,}440{,}000$). Only hot (expanding) gas is plotted on the left-hand side, and cool (contracting) gas on the right-hand side, thus illustrating the change of $T_{\rm exc}$ through the light cycle. Between phases 0.90 and 0.10, the spectrum is double-lined, and both hot and cool gas can be seen.

TABLE 3

$^{12}C^{16}O$ $\Delta v=3$ EXCITATION TEMPERATURES

JD (2,440,000+)	Excitation Temperature	
2790.3	2800	
2792.3	2700	
2843.2	2700	
2876.1	2600	
2881.1	2400	
2910.1	2500	
2937.1	2400	3900
2985.9	2200	3600
3007.6	2200:	3500
3050.7	3200	
3088.6	3200	
3118.4	3100	
3155.3	2900	
3183.3	2800	
3194.2	2700	
3231.2	2700	
3294.7	2700	
3319.0	2500	
3381.7	2400:	3500
3418.6	3400	
3441.5	3300	
3478.3	3300	
3656.0	2700	
3798.6	2200:	3500
3851.5	3400	
3913.2	3100	
3924.2	3000	

0.70. An order of magnitude estimate of the actual CO column density can be made by assuming that the weakest ($\log W_\lambda/\lambda < -5.3$) lines are on the linear portion of the curve of growth. An isothermal, plane-parallel model fitted to the weak line portion of the observed curves of growth gives the CO column density scale of the Figure 6 ordinate.

An estimate of the total column density through the CO $\Delta v=3$ line-forming region can be made from the CO column density. Assuming that (1) C/O=0.9 (Scalo and Ross 1976), (2) carbon has been enriched and the oxygen not depleted (Thompson 1977), (3) O/H has the solar value, and (4) all the carbon is associated into CO, then the total column mass is $mL \approx 4200$ g cm^{-2} when $\log NL(^{12}CO)=24.3$. Assuming all the hydrogen is associated into H_2 near minimum light, $\log NL(H_2) \approx 27.1$. This column should produce strong quadrupole H_2 lines in the 2 μm region, and these are indeed prominent in the χ Cygni spectra around minimum light (Hall *et al.* 1982).

The curves of growth (Fig. 4) show significant changes in the shape of the strong line portion with phase. The shape change in the curves of growth can be summarized as a well-defined flat portion before phases 0.25, straightening until nearly linear at later phases. The curve of growth retains near linearity even during the infalling phases past phase 1.0. The difference in maximum-postmaximum and minimum-premaximum shapes of the curves of growth must result from a phase-dependent change in the line forming regions of the atmosphere.

IV. THE FIRST OVERTONE ($\Delta v=2$) BANDS

Large numbers of strong CO lines are present in the first overtone of χ Cygni. At wavenumbers smaller than that of the 4–2 band head (4250 cm^{-1}), the spectrum

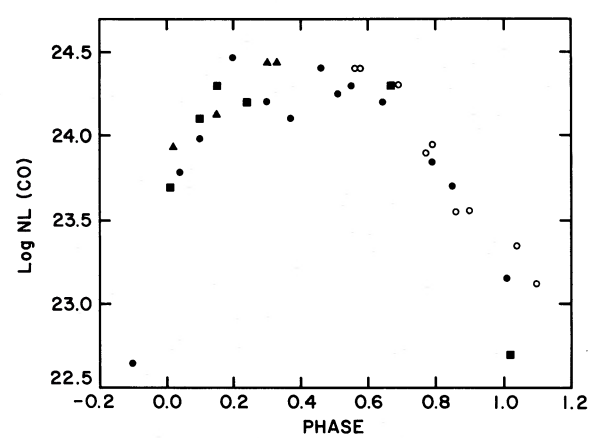

FIG. 6.—Log of the total CO column density in the photosphere as a function of phase. Symbols and plotting convention at double-line phases are as in Fig. 5.

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1982ApJ...252..697H

becomes a jumble of overlapping lines. All Earth-based observers suffer from the additional complication that only the bluest 200 cm^{-1} are present before the sequence is obscured by the blue wing of the opaque 2.7 μm telluric water band. As a result, most of the information available in the CO first overtone bands of χ Cygni comes from the *R*-branches of the 2–0 and 3–1 bands. In the 2–0 band, most of the low excitation lines from roughly $J''=0$ to 24 and most of the high excitation lines from about $J''=79$ to 87 are unblended. In the 3–1 band, a series of moderate excitation lines from $J''=30$ to 40 is well placed. These three sets of lines appear to be formed through different ranges of continuum optical depth, with each group of lines accenting one of the multiple velocity components present more clearly than the others.

a) The 800 K Component

Figure 7 shows a section of the spectrum near 4325 cm^{-1} which contains clean high and low excitation 2–0 lines. The low excitation lines have several notable properties. Near minimum light, the cores have depths near 100%, while the strongest photospheric $\Delta v=3$ CO lines have ~75% central depths. Assuming that the depths of the strongest $\Delta v=3$ lines are governed by the difference in temperature between the upper and lower photospheric layers, the cores of the low excitation 2–0 lines are formed above the boundary layer for the photospheric lines. Furthermore, measurements of the radial velocities of the deepest point in the low excitation 2–0 lines show that they do not exhibit the periodic change in velocity of either the second overtone lines or the high excitation 2–0 lines. The velocity difference between the high and low excitation 2–0 lines is probably a characteristic common to all late-type Mira spectra. Maillard (1974) first noted this effect in R Leo, and we have seen it in the spectra of about a dozen other late-type Miras.

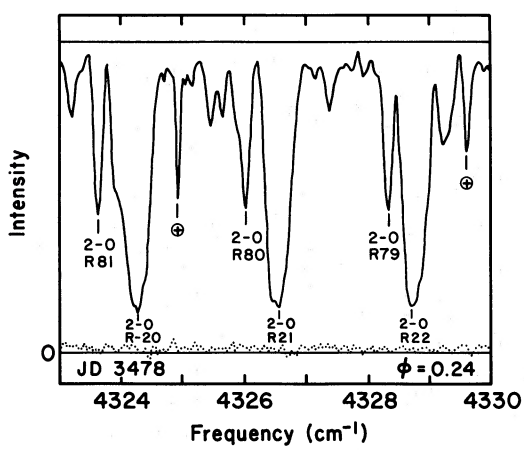

FIG. 7.—Section of χ Cyg CO first overtone spectrum taken on JD 2,443,478 (phase 0.24). The line above the spectrum is the continuum. The difference between forward and reverse scans (*dotted line*) gives an estimate of the noise. This section of spectrum contains some strong, low excitation CO 2–0 lines (R20, R21, R22) that indicate an 800 K component. These lines have a different radial velocity than the high excitation, photospheric 2–0 R81, R80, R79 lines.

In Figure 8, composite profiles for the low excitation 2–0 lines are displayed for representative phases in several light cycles. Velocities derived from composite low excitation 2–0 line cores are presented in Table 2. The nonphotospheric component in the cores of the 2–0 lines varies but not in a way obviously related to the light cycle. During the interval from JD 2,442,790 until 2,443,470, the low J'' 2–0 line cores were dominated by the nonphotospheric component. During this period, the velocities of these lines did not change by more than ±2 km s^{-1}, with this change probably resulting from blending and not from an actual change of velocity. Starting with the JD 2,443,656 spectrum, the velocity of the 2–0 low J'' lines appeared to change from the previous constant value. In the composite profiles, the photospheric part of the 2–0 low J'' line profile increasingly dominates the total line profile after this time. Plots of a spectrum taken during instrumental development before the start of the time series on JD 2,442,570 resemble the spectrum on JD 2,443,799.

The low excitation-high excitation velocity difference measured by Hinkle, Lambert, and Snell (1976) from an even older spectrum (JD 2,441,827) of χ Cygni shows that the low J'' nonphotospheric component was also weak at this time. We conclude that the observed changes in velocity of the low excitation lines probably result from the increasing importance of the photospheric contribution to the line profile and do not indicate a change in velocity of the low excitation gas. Evidently, changes in the amount of gas contributing to the low excitation line cores occur, but these changes are not simply correlated with the 407^d stellar period.

From the average CO $\Delta v=2$ profiles (Fig. 8), the futility of attempting to separate the various components present in the low excitation lines is apparent. Nevertheless, for a few postmaximum spectra, the nonphotospheric low excitation lines are partially resolved from the same transition in the photospheric spectrum, and it is possible to crudely measure the excitation temperature and the CO column density. The best spectra for this use are the four spectra taken between 1977 August 26 and November 30 of phases 0.03 to 0.27. The depths of the 2–0 low J'' lines were measured at the radial velocity found from the average line profile plots. The core of the low J'' component appears ~8 km s^{-1} across, so lines having cores at velocities differing by more than ±4 km s^{-1} from the average profile low J'' velocity were assumed blended with telluric or photospheric lines from a different transition and not used.

The photospheric contribution to the line depths was removed by assuming that the underlying low J'' photo-

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1982ApJ...252..697H

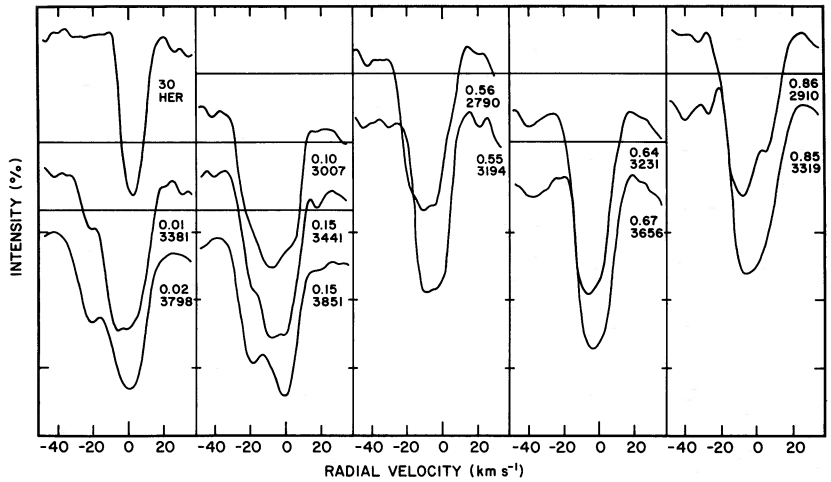

FIG. 8.—Average 2–0 low excitation profiles. Numbers beside each profile are the phase (*above*) and JD (−2,440,000). The line above each profile is the continuum, and *y*-axis ticks mark zero intensity. Five different phases are shown for several different light cycles to emphasize the large cycle-to-cycle differences. Note weakening at phase 0.10 between JD 3007 and 3851. 30 Her is shown as a comparison.

spheric lines are saturated at a constant depth. From the 1978 October through 1979 February observations of phase 0.02 through 0.30, the low J'' component is weak enough that the cores of the photospheric profile are obvious. Assuming symmetric photospheric line profiles, the photospheric contribution to the observed profiles was removed, and a linear equivalent width versus central depth relation derived.

Forcing the high and low excitation lines to coincide on the curve of growth, an excitation temperature of 800 ± 100 K was found for each of the four spectra. Using a plane-parallel, isothermal model, the curve of growth can be fitted with a microturbulent line width of 1.4 ± 0.4 km s^{-1} and a CO column density of $\log NL(^{12}\mathrm{CO}) = 20.0 \pm 0.03$. These results are insensitive to the assumptions made measuring the lines. Making the same assumptions concerning the number of H_2 to CO molecules as for the photospheric second overtone lines, $\log NL(H_2) \approx 22.8$. The weakening of the 800 K lines in 1978 implies a decrease in $\log NL$ of ~ 0.8, providing that the excitation temperature remained constant.

b) *The Photospheric Component*

Up to ~ 100 unblended high excitation first overtone lines can be found on the best spectra, although typically 25 or fewer lines are available. From the best spectra, information on the velocities as a function of excitation potential can be retrieved. Considerable dependence of the velocity on excitation potential exists, probably due to blending with the 800 K component, with the low and high excitation lines differing by as much as 6 km s^{-1}. In order to remove a functional

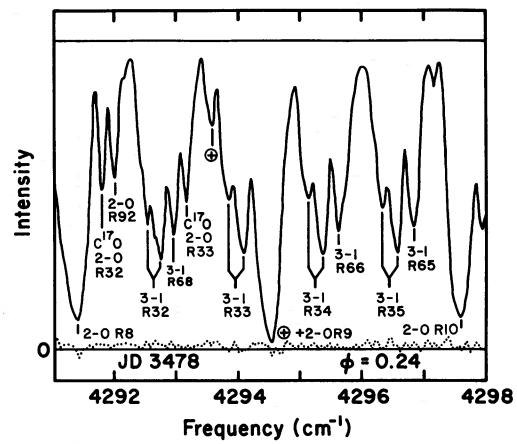

FIG. 9.—Another section of the spectrum shown in Fig. 7. This region contains the moderate excitation 3–1 lines (R32, R33, R34, R35) near the 3–1 band head. These lines are doubled, with one component at the photospheric velocity and another indicating infalling gas.

dependence on excitation from the velocities, the clean 2–0 R79, 80, 81, and 82 lines were combined into an averaged profile, and the velocity measured from the minimum of this profile. The resulting velocities are listed in Table 2. While the behavior of these 2–0 lines is nearly identical to that of the second overtone lines, slight velocity differences exist. In particular, the high excitation 2–0 lines double later in phase than the $\Delta v = 3$ lines, presumably because of the increase in continuous opacity from 1.6 to 2.3 μm.

Blending makes the 3–1 lines considerably more difficult to analyze than the 2–0 lines. Of the 64 3–1 lines

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1982ApJ...252..697H

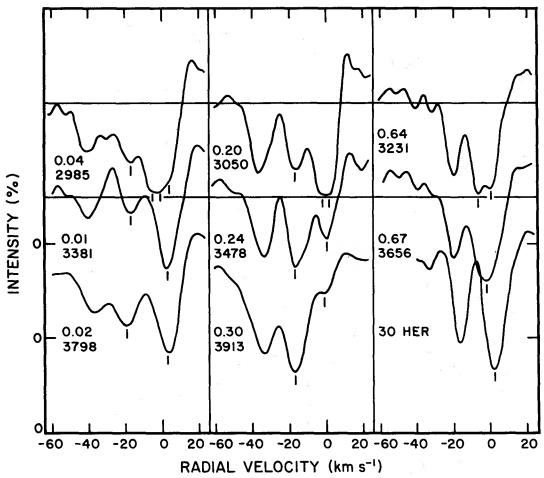

FIG. 10.—Average CO 3–1 moderate excitation line profiles. Format is the same as for Fig. 8. The line below the profiles indicates suspected velocity components. Average line profiles are complicated by a high excitation 3–1 line near each moderate excitation line selected. This is conspicuous in the 30 Her comparison profile, as well as from comparison of JD 2,443,478 profile with R34 and R35 lines in Fig. 9. Note the systematic weakening from one cycle to the next of the infalling (0 km s^{-1}) component at phases 0.2–0.3.

present between the $^{13}C^{16}O$ 2–0 band head at 4265 cm^{-1} and the 3–1 band head at 4305 cm^{-1}, most are blended either together or with strong low excitation 2–0 lines. Only five 3–1 lines can be found which are of moderate excitation and remain unblended throughout the light cycle: R33, 34, 35, 37, and 38. The spectral region centered on R33 and 34 is shown in Figure 9.

Average line profiles compiled from the five unblended 3–1 lines are shown in Figure 10 for a number of phases in different cycles. The presence of a higher excitation 3–1 line (R65, 66, and 68 in Fig. 9) near each of the selected 3–1 lines complicates the interpretation of the average line profiles, but comparing the line profile in a low amplitude, variable M giant (30=g Her) with the χ Cyg profiles, it is obvious that χ Cyg has extra components present. In fact, the presence of extra components is conspicuous, if hard to interpret from plots of the spectra (Fig. 9), when it is noticed that the region near the 3–1 band head has at least twice as many lines as it should!

One of the 3–1 components always matches the 2–0 high excitation velocity. The other component(s) present is either at the velocity of the low J'' 2–0 lines or at positively shifted (infalling) velocities relative to the Dickinson *et al.* (1978) center of mass velocity. Considerable cycle-to-cycle variation is conspicuous. In some cycles, the infalling line is single; in others, multiple components seem likely. In most cycles, the infalling gas could be followed until about phase 0.5, when the photospheric and infalling velocities become equal, blending the velocity components together. Velocities of components that could be well established from the time series are listed in Table 2.

From the best-resolved, highest signal-to-noise spectra, the central depths of all lines appearing at the infalling velocity were measured, and while insufficient lines were present for curve of growth analysis, the lines available indicate a temperature between the photosphere (≥2000 K) and the 800 K component.

V. THE FUNDAMENTAL ($\Delta v=1$) BANDS

By virtue of large oscillator strengths, 50 times those of the $\Delta v=2$ transitions, the 1–0 band is the most sensitive indicator of cool circumstellar CO. Bernat *et al.* (1979) reported that CO fundamental lines formed in the circumstellar envelope of α Ori are present with considerable strength in the 4.6 μm spectra. These lines are of particular interest since collisional processes dominate radiative ones for the CO 1–0 lines, allowing, in principle, a direct measurement of the kinetic temperature of the shell.

We obtained five spectra of the 4.6 μm region of χ Cygni (Table 1B) primarily to search for the circumstellar CO lines. The telluric spectrum contains heavily saturated low excitation CO lines that are opaque in the cores. In order to observe the circumstellar lines, which have comparable or lower excitation temperature than the telluric lines, spectra must be obtained when the combined stellar radial velocity and the Earth's orbital velocity provide a sufficient Doppler shift to separate

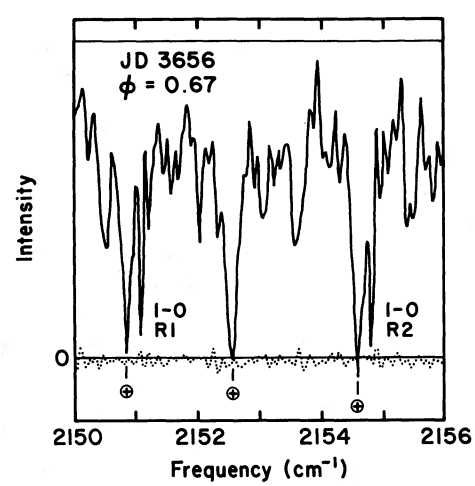

FIG. 11.—Section of the CO fundamental spectrum near 4.6 μm for χ Cyg on JD 2,443,656 (phase 0.67). Continuum and noise level indicated as in Fig. 7. Note the great depth of the circumstellar 1–0 R1 and R2 lines compared to the other stellar features (telluric lines are indicated).

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1982ApJ...252..697H

TABLE 4

CO $\Delta v = 1$ RADIAL VELOCITIES

Date	Circumstellar	Photosphere
1977 Feb 9.....	-15.0 ± 0.2	-2:
1977 Oct 3.....	...	$+1$
1978 Feb 17	-15.6 ± 0.2	0, -12
1978 May 22 ...	-15.2 ± 0.2	-1.5
1978 Oct 18	...	$+3.2$
Mean	-15.3 ± 0.2	

the telluric and circumstellar CO lines. Since the circumstellar maser lines of Miras are known to have complex structure, we observed χ Cyg at both the shift for the "normal" expanding circumstellar absorption lines and at the opposite shift to check for unexpected features. Only the "normal" blueshifted component was detected; it is a conspicuous feature in the spectra (Fig. 11). The three spectra obtained at the correct velocity shift to show circumstellar lines were taken with a filter that only transmitted a few *R*-branch lines. These lines are sufficient to measure the radial velocity but do not allow the excitation temperature to be determined accurately.

The radial velocities of the $^{12}C^{16}O$ 1–0 circumstellar lines are presented in Table 4. The velocities were determined from the minima of about eight lines, generally R1 through R8. No velocity variation with time is present. The heliocentric velocity of -15.3 km s^{-1} ± 0.2 gives an implied expansion velocity of 7.8 km s^{-1} if a stellar center of mass velocity of -7.5 ± 0.6 km s^{-1} is adopted (Dickinson *et al.* 1978). The excitation temperature is 300 ± 200 K. (High resolution [$\lambda/\Delta\lambda = 100{,}000$] spectra taken in 1980 reveal that two velocity components are present in the χ Cyg circumstellar CO lines. One is at -17.6 km s^{-1} at 60 ± 20 K, and the other at -13.9 km s^{-1} and 300 ± 200 K.)

It is interesting to attempt to identify other circumstellar and photospheric components that contribute to the 4.6 μm χ Cygni spectra. The maximum resolution attainable at the time the observations were taken was ~ 0.06 cm^{-1}, which corresponds to 10 km s^{-1} in the 4.6 μm region. At this resolution, blending between the various photospheric and 800 K components is a severe problem. Average line profiles were computed for a set of relatively unblended lines. The resulting velocities are given in Table 4. The 800 K component is not identifiable in the average profiles, and the velocities indicate that the photospheric spectrum seems to be dominated by the infalling component. The photospheric lines are also relatively shallow, having a maximum depth of 60%; this makes the deep circumstellar lines conspicuous in the spectra (Fig. 11).

VI. DISCUSSION

To summarize the preceding section, four components have been identified in the spectra.

1. Second overtone CO lines and the high excitation component of the first overtone lines. These lines have an excitation temperature ranging from 4000 to 2200 K and a discontinuous velocity curve of nearly 30 km s^{-1} amplitude. Both properties indicate that the lines are formed in a pulsating photospheric region traversed by a shock front near maximum light.
2. First overtone lines with infall velocities. This group of lines persists to at least minimum light and may well be present throughout the entire cycle.
3. Low excitation first overtone lines with stationary velocities. These lines have an 800 K excitation temperature, a velocity similar to the center of mass velocity, and equivalent widths that vary on a longer time scale than the pulsation period. This component is suggested to originate in the inner region of the circumstellar shell.
4. Low excitation fundamental lines. These lines indicate gas expanding at 7.8 km s^{-1} at ~ 300 K excitation temperature. These lines are formed in the expanding circumstellar shell.

In the following subsections, the information from the various spectral features is combined to produce an empirical picture of the atmosphere of χ Cygni.

a) *Photosphere*

In Figure 12, the velocities of the second overtone CO lines are shown as a function of visual phase throughout

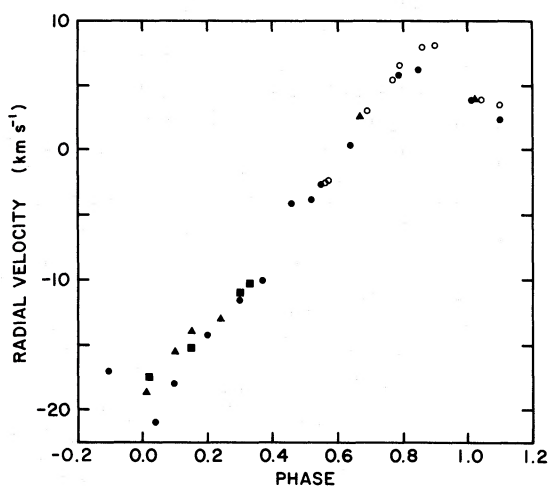

FIG. 12.—CO $\Delta v = 3$ velocities shown in Fig. 2 plotted as a function of phase. Symbols and convention for plotting components at double-lined phases are the same as for Fig. 5. The center of mass velocity of χ Cyg is -7.5 ± 0.6 km s^{-1}.

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1982ApJ...252..697H

a complete pulsation. Velocity curves of this shape are the result of radial stellar pulsation driving compression waves which develop shocks before emergence into the photosphere. Several classes of pulsating stars, including the Population II Cepheids (Wallerstein 1959) and the RV Tauri stars (Preston 1964), have similar velocity curves. Lines in the visual spectrum of LPVs do not, however, exhibit this velocity curve, and direct kinematic evidence of the pulsating photosphere is seen only in the infrared. Initial observation of this type photospheric velocity curve for a Mira (R Leo) is presented by Hinkle (1978). Preliminary discussion of χ Cygni is given by Hall, Hinkle, and Ridgway (1979). We review and update these discussions in light of the present observations.

The first indication of the emergence of the shock into the χ Cygni atmosphere is probably the brightening of the star after minimum light near phase -0.40. Between phase -0.40 and -0.30, the shock propagates through the infrared (1.6 μm) continuum-forming layers. The infrared line-forming layers are first affected at phase -0.30 when the CO column density (Fig. 6) begins to decrease. By about phase -0.10, the expanding gas behind the shock has cooled to the point where a detectable column density of CO can form. From phases -0.10 to 0.15, the CO continues to associate behind the shock zone, while CO is simultaneously being dissociated by the advancing shock. The shock's velocity amplitude (>30 km s^{-1}) is sufficient to dissociate the CO, a phenomenon consistent with the existence of hydrogen emission lines (Merrill 1953; Willson 1976), since the ionization of hydrogen requires 13.6 eV, while the dissociation of CO requires 11.1 eV. By phase 0.15, the CO column density reaches its plateau value, and from then until phase 0.7, it remains constant to within our accuracy of measurement. During this interval, the line-forming layer expands, stops, and falls back toward the stellar surface under a constant inward acceleration of 0.098 ± 0.005 cm s^{-2}. The CO column density drops an order of magnitude by phase 0.85 and another by phase 1.1. The observed acceleration of the layer changes at phase 0.85, and the line forming layer reaches an equilibrium infall value of 7.5 km s^{-1} which is maintained by a small amount of gas until at least minimum light.

The principal differences between the model for the atmosphere-shock interaction presented here and the model discussed in previous papers by the authors is that we now recognize that the shock has already passed through much of the photosphere when the CO second overtone spectrum becomes double lined. Consequently, we now believe that the turnover of the velocity curve after phase 0.85 results from the shock having passed through enough of the photosphere so that a higher atmospheric region provides the residual layer which is moving at a different velocity.

Two minor details of the spectra fall into place with this scenario for the passage of the shock through the atmosphere. First, weak second overtone CO emission lines develop after phase -0.4 and last until about phase -0.1 (Fig. 1). We suggest that these lines result from the recombination of CO behind the shock. The change from emission to absorption lines occurs when the absorption component of associated CO seen against the photosphere overtakes the strength of the emission from the recombining CO. Second, Figure 12 indicates excellent agreement of CO second overtone velocities from different cycles between phases 0.15 and 0.70. These are the phases when the column density is constant. The velocities do not agree as well from cycle to cycle before phase 0.15, as the CO is recombining, and after phase 0.70, as the CO is dissociating. During these periods, the CO velocities are probably influenced by atmosphere inhomogeneities and other differences from one cycle to another which affect either the rates of molecular association and recombination or the shock propagation.

From the basic picture of the pulsation outlined above and from measurements of the velocity, temperature, and column density as a function of phase, it is possible to deduce a considerable amount of information concerning the atmospheric structure.

Our discussion of the stellar atmosphere begins with the stellar radius. Values in the literature for Mira radii range over an order of magnitude from 180 $R_\odot$ (Labeyrie *et al.* 1977) to 1800 $R_\odot$ (Nather and Wild 1973). The results of Labeyrie *et al.* also indicate a strongly wavelength-dependent radius. Recently, angular diameters have been measured for several Miras including χ Cygni (Christou and Worden 1980), but large uncertainties in the distance and the variation of radius with wavelength make these difficult to interpret.

An estimate of the photospheric radius can be made using the relation

$$L = R^2 T^4,$$

where the luminosity L, effective temperature T, and radius R are in solar units. For the luminosity, we adopt $M_{\rm bol} = -5 \pm 1$ for all Miras at mean light (Smak 1966; Cahn and Wyatt 1978). The uncertainty in the luminosity is comparable to the uncertainty in distance to an individual Mira. In order to find the radius of the photosphere unaffected by the presence of the three overlying layers, the CO excitation temperature was used. Under photospheric conditions, collisions dominate the radiative processes in CO, so the CO excitation temperature is the kinetic temperature. Model atmosphere calculations for nonpulsating M giants show that moderate strength lines ($\log W_\lambda/\lambda \approx -5$) such as those

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1982ApJ...252..697H

used to find the CO excitation temperatures in χ Cyg have excitation temperatures $\sim$400 K cooler than the effective temperature (Dominy, Hinkle, and Lambert 1982). A mean excitation temperature of 3000 K (Fig. 5) plus the 400 K correction, combined with the 7000 $L_\odot$ luminosity, gives a radius of 240 $R_\odot$, i.e., 1.7×10^{13} cm. This value is near the previous lower estimates of Mira radii. In the case of R Leo, the hotter 3500 K excitation temperature (Hinkle 1978) produces $R_* \approx 180\ R_\odot$.

Considerable uncertainty is involved in these radius estimates. The uncertainty in the bolometric magnitude corresponds to $\pm 110\ R_\odot$ in χ Cygni. The CO excitation temperature in Miras is probably not related to the radius in the same way that it is for the nonpulsating M giants. The range of excitation temperature in a typical M giant model atmosphere (Johnson, Bernat, and Krupp 1980) is 1000 K, which corresponds to an uncertainty of $\pm 50\ R_\odot$ in the stellar radius estimate.

An independent estimate of radius can be made from the angular diameter of 23 milli-arcseconds measured at 6000 Å (Christou and Worden 1980). The distance modulus can be estimated from I band photometry (Barnes 1970). The observed value of $m_I = -0.1$ at maximum light for χ Cygni (Barnes 1973) gives a distance of 132 ± 25 (formal error) pc. The radius is then 316 ± 60 $R_\odot$. This does not provide an improvement on the above estimate because a factor of 2 wavelength dependence in the angular diameters of Miras seems indicated

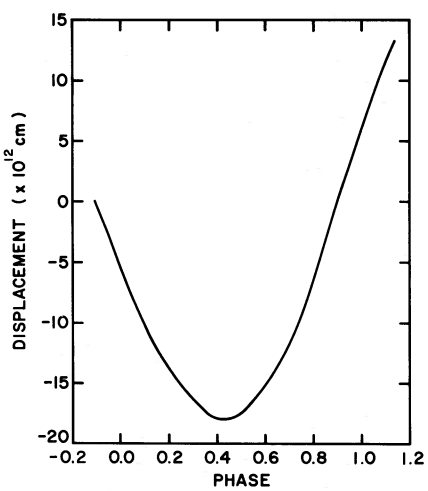

FIG. 13.—Displacement as a function of phase found from integrating the CO $\Delta v = 3$ velocity curve (Fig. 12), assuming a center of mass velocity of -7.5 km s^{-1}. Zero displacement is assumed for the expanding gas at phase 0.90. The zero displacement at phase 0.90 for the falling gas results from this choice. Note that the column density of CO is increasing until phase 0.15 and decreasing after phase 0.70.

(Labeyrie *et al.* 1977). We adopt the 240 $R_\odot$ value in the subsequent discussion.

The constant acceleration of the photospheric material over much of the cycle (Fig. 12) suggests that the material moves ballistically during this period. If, however, the constant acceleration of 0.098 cm s^{-2} is identified with the surface gravity at a radius of 1.7×10^{13} cm, then the implied stellar mass is 0.2 $M_\odot$ for χ Cygni which is absurdly small. The simple ballistic interpretation is, in any case, ruled out because the displacement of 1.7×10^{13} cm obtained by integrating over the constant acceleration phase with a center of mass velocity of -7.6 km s^{-1} and a limb effect correction of 1.31 (Parsons 1972) implies a factor of 4 change in surface gravity.

The thickness of the CO $\Delta v = 3$ line-forming region can be estimated from the time it takes the rising shock to traverse the falling line-forming region. A complication arises in that the velocity of the falling gas above the shock is directly measurable, but the shock velocity is not. The expanding gas behind the shock is seen in χ Cygni after phase -0.10. The shock velocity cannot differ from this velocity by more than the velocity of sound behind the shock. The measured temperature indicates that the sound velocity is $\sim$5 km s^{-1}. Prior to phase -0.10, weak CO $\Delta v = 3$ emission lines are seen (Fig. 1) at the expected velocity of gas behind the shock. These lines are $\sim$7 km s^{-1} from the center of mass velocity at phase -0.20. At phase -0.10, the absorption lines display an expansion rate of $\sim$11 km s^{-1}. Assuming the shock is moving outward at 7 km s^{-1} at phase -0.2 and 11 km s^{-1} at phase -0.1 with a 1.31 limb effect correction factor, the shock covers $\sim 8\times10^{12}$ cm between phases -0.30 and -0.10. In the same time, the falling gas travels 1.2×10^{13} cm (Fig. 13), indicating a total thickness of the linearly accelerated part of the atmosphere of $\sim 2\times10^{13}$ cm. From the column density versus phase diagram (Fig. 6), the column density has fallen by $1/e$ at phase -0.20, and the corresponding scale height is $\sim 8\times10^{12}$ cm for the infalling, premaximum atmosphere. While the shock velocity is poorly known, the measured velocity of infall is large, resulting in an uncertainty in the scale height of less than $\pm 3\times 10^{12}$ cm. The scale height of half the radius implies that spherical geometry cannot be ignored since the total height of the atmosphere undergoing the pulsation is about equal to the radius.

Knowing the thickness of the line-forming region, the mass of the atmosphere can be found. From this, the energy involved in the pulsation can be computed. Assuming that $mL = 4200$ gm cm^{-2} and that the mass is distributed exponentially through the line-forming region, the mass of the stellar atmosphere is $\sim 2.5\times10^{31}$ g. As the disturbance propagates out through this layer, it

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1982ApJ...252..697H

both heats and outwardly accelerates it. The observed dissociation of CO during this phase implies that H_2 must be dissociated, a process which requires $\sim 3\times 10^{43}$ ergs and is an order of magnitude greater than the $2\text{–}4\times 10^{42}$ ergs required to simply heat the layer from 2000 to 4000 K. Averaged over the 408^d period, this is $\sim 3\%$ of the luminosity. If most of the atomic hydrogen is ionized by the shock, then the energy is raised by a factor of 3. The 13.6 eV ionization energy of hydrogen is similar enough to the 11.1 eV dissociation energy of CO that the higher estimate is probably appropriate. Relative to the center of mass velocity of -7.5 km s^{-1}, the speeds of the layer before and after acceleration are essentially the same so that there is no change in the kinetic energy, which is $\sim 10^{43}$ ergs for an observed speed of 10 km s^{-1} and a limb correction factor of 1.3. However, the momentum transferred to the layer is $\sim 3\times 10^{37}$ gm cm s^{-1}, which, averaged over the period, is three orders of magnitude greater than L/c. Although the majority of the material remains bound to the star, selective momentum transfer to a small fraction of it could play a major role in the mass loss process.

The present observations thus provide the first comprehensive phase coverage of the kinematics and energetics of the photospheric pulsation in a long period variable. As such, they should permit further evaluation of theoretical models which invoke some general, subphotospheric oscillation to drive the observed variability (Hill and Willson 1979; Wood 1979). Such models have the potential for relating the observed parameters of the variability to surface gravity and stellar mass when the controversy as to whether the stellar oscillation is in a fundamental or overtone mode is resolved.

b) *Extended Atmospheric Layers*

In addition to the photospheric layers discussed in the preceding section, the CO observations reveal three other regimes. These are (1) cool ($T\approx 300$ K) gas expanding at 7.5 km s^{-1} relative to the center of mass, (2) a hotter ($T\approx 800$ K) stationary shell, and (3) still hotter ($T\approx 1500$ K) gas infalling at ~ 8 km s^{-1}. In this section, we assume the progression of decreasing temperature corresponds to increasing distance from the photosphere and discuss the implications for the structure of the circumstellar envelope and its relation to visible and microwave observations.

The coolest component corresponds to the classical expanding circumstellar shell. The observed velocity of -15.3 ± 0.2 km s^{-1} is in excellent agreement with the values of -16.5 km s^{-1} measured for visual resonance lines (Wallerstein 1975) and -15.0 ± 0.7 km s^{-1} for the blue edge of the thermal profile of the $J=2\text{–}1$ ground vibrational state transition of SiO (Dickinson *et al.* 1978). If the ~ 200 K temperature is determined by collisions with dust whose temperature is in turn governed by gray body absorption and reemission of a 3200 K radiation field, then this material is at a typical distance of 100 stellar radii from the photosphere. This distance of $\sim 1.5\times 10^{15}$ cm and the expansion velocity of 7.8 km s^{-1} then imply expansion time scales ≤ 100 yr. Better quality observations of lines with J-values up to at least 20 in the CO 1–0 band are required to accurately specify the circumstellar temperature and column density and thus the mass loss rate.

The 800 K component appears to be time variable on a scale significantly longer than the optical period. It is entirely absent from an exploratory spectrum taken near maximum light on 1975 June 6 (JD 2,442,576), yet it is pronounced in early spectra of the present series. However, it has steadily weakened over the three cycles and is barely distinguishable in the most recent spectra. Throughout this period, the gas has remained at velocities within a few km s^{-1} of the center of mass velocity (Table 2). If the 800 K temperature is determined by absorption, collisional thermalization, and subsequent reradiation by gas in the near infrared ($1.5<\lambda[\mu\text{m}]<10$), then the layer must be of the order of 10 stellar radii from the photosphere. If it is instead heated by dissipation of the outwardly propagating waves from the stellar pulsation, then it may be substantially further out. A radius $\sim 10\ R_*$ (1.7×10^{14} cm) and a column density of $\log NL(H_2)>22.8$ (as measured on JD 2,443,478) implies a total mass $>10^{-5}\ M_\odot$. The observed line width of 8 km s^{-1} is at least 4 times the width required to fit the curves of growth, implying there is considerable turbulence in the 800 K layer.

A plausible hypothesis is that the 800 K layer represents gas ejected from the star, probably by a particularly violent oscillation, and supported by dissipation of shock energy. As such, it could provide a source from which mass loss driven by radiation pressure on dust may originate. The 800 K temperature is in the right range for condensation of dust (Gilman 1969), and the lower limit on the mass is large enough to produce a reservoir for the canonical $10^{-6}\ M_\odot$ yr^{-1} mass loss rate (Gehrz and Woolf 1971; Hagen 1978). Such a layer is present in the 10 Miras for which we have obtained infrared spectra and has been identified in R Leo (Hinkle 1978). It may be present in most late-type stars as a critical step in the mass loss process but is easily distinguished only when the photospheric lines are shifted away from the 800 K layer lines by the Mira's pulsation.

The infalling gas revealed by CO 3–1 lines of moderate excitation is difficult to separate from the other components. The velocities of these lines are plotted in Figure 14, and it is clear that at premaximum phases confusion exists between the lines from infalling gas

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1982ApJ...252..697H

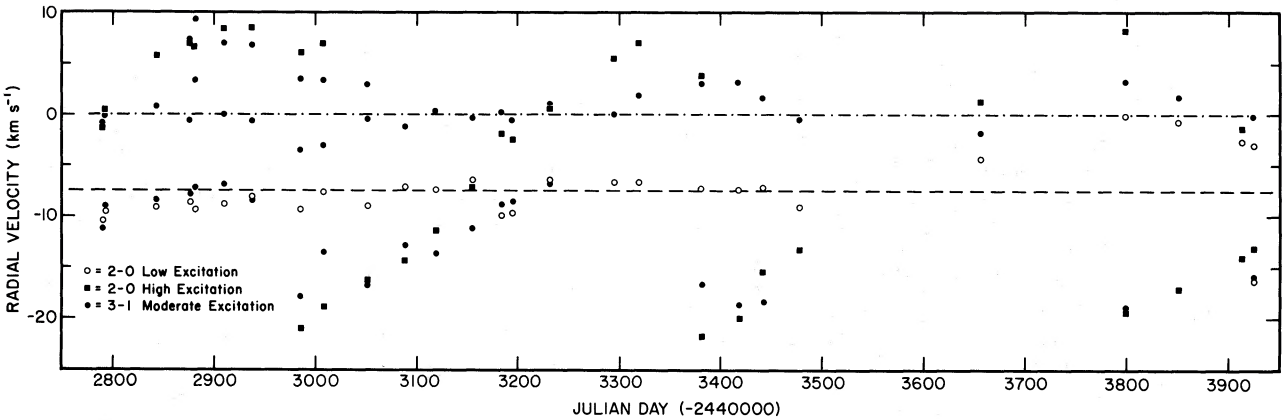

FIG. 14.—CO first overtone velocities for three different groups of lines plotted as a function of JD number. *Dashed line*: center of mass velocity (Dickinson *et al.* 1978). *Dot-dashed line*: mean visual region "photospheric" velocity given by Wallerstein (1975).

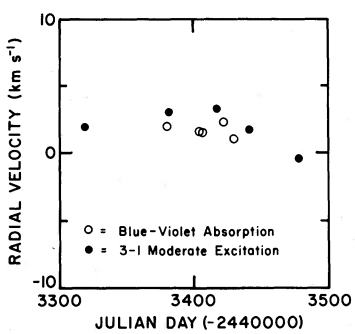

FIG. 15.—Average velocities of the $^{12}C^{16}O$ 3–1 moderate excitation lines (*filled symbols*) and blue-violet region velocities measured during the same time interval by Pierce, Willson, and Beavers (1979, *open symbols*).

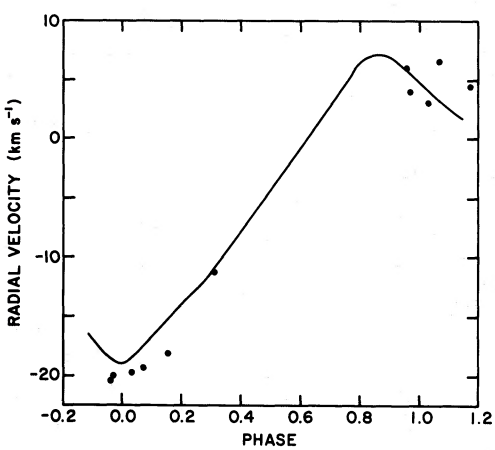

FIG. 16.—Maehara's (1971) atomic line velocities from the 8000 Å region plotted as a function of phase. *Solid line*: CO $\Delta v = 3$ mean velocity curve from Fig. 12.

near 0 km s^{-1}, the stationary 800 K gas, and the photosphere. However, infalling gas certainly exists until phase 0.5 and may persist through the entire light cycle. From Figure 10, the column of infalling gas appears to decrease over the period of observation in the same way as the 800 K component. This suggests they are related and that the infalling component represents material draining back down to the photosphere from the 800 K layer.

It seems likely that the infalling region is the source of the visible spectrum. Visual radial velocities were measured by Pierce, Willson, and Beavers (1979) during the same light cycles as some of the infrared velocities. Velocities of the visual, absorption lines are plotted in Figure 15 and are in good agreement with the infalling CO component. The visual spectrum does not have nearly as pronounced a phase-dependent velocity change as seen in the 1.6 μm photospheric lines, indicating the visual lines are formed above the 1.6 μm line-forming region. If this is the case, the scale height of the infrared-photospheric region implies that a factor of at least 2 difference in the angular diameters measured in the visual, and the 1.6 μm infrared is expected (so long as center to limb effects are similar at the two wavelengths).

The opacity that prevents seeing deeper than the 0 km s^{-1} infalling gas in the visual must be strongly wavelength-dependent since Maehara (1971) found that near 8000 Å doubled CN and atomic lines are visible in χ Cygni near maximum light. Maehara's velocities and a mean CO $\Delta v = 3$ velocity curve are shown in Figure 16. The velocities measured in the 8000 Å region are a good fit on the CO velocity curve. The 8000 Å line doubling is undoubtedly a manifestation of the same shock which produces the photospheric CO line doubling. Due to the absence of premaximum observations, Maehara's observations have insufficient phase coverage to show that

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1982ApJ...252..697H

the velocities form a single discontinuous curve, and the velocities are shown as separate components in his paper.

The 800 K shell seems the only plausible source of SiO maser emission from the $v=1$, $J=2\rightarrow 1$ (Kaifu *et al.* 1975) and $v=1$, $J=1\rightarrow 0$ (Snyder and Buhl 1975) transitions in χ Cygni. Both are centered at -9 km s^{-1} (heliocentric) and extend from -7 to -11 km s^{-1}. This would suggest that the SiO maser emission arises in localized clumps within a circumstellar cloud of diameter $\sim 3\times 10^{14}$ cm, a value consistent with VLBI determinations $\sim 8\times 10^{13}$ cm for R Cas and $\sim 10^{15}$ cm for VX Sgr (Moran *et al.* 1979). An important test of this hypothesis would be a search for long-term variations in SiO maser characteristics correlated with the formation and dissipation of the 800 K layer.

VII. SUMMARY

Analysis of CO vibration-rotation lines in a time series of high resolution infrared spectra of χ Cygni has provided detailed information about both the photospheric kinematics and energetics associated with the optical variability and the nature of the complex circumstellar envelope. The overall picture which results is consistent with visible and microwave observations and specifies the regions where low excitation optical absorption lines and the SiO maser features originate.

Observations of the CO $\Delta v=3$ bands around 1.6 μm show that the atmospheric structure is governed by a general stellar pulsation which propagates through the stellar photosphere during the premaximum and maximum phases. A shock front forms between gas infalling from the previous cycle and material accelerated by the new wave. At this shock interface, the gas is heated from ~ 2000 to >4000 K and molecular species are dissociated; atomic hydrogen is probably predominantly ionized. The energy involved in these processes is $\sim 10\%$ of the stellar luminosity if averaged over the period, and the outward momentum flux is about three orders of magnitude greater than L/c. After shock has propagated through the photosphere, the material is uniformly accelerated inwardly and cools monotonically for most of the cycle (phases 0.1 to 0.8). Simple arguments indicate the stars are much more extended than normal M giants with a radius to the 1.6 μm CO-forming layer of ~ 250 $R_\odot$ and a similar thickness for the layer; the stellar pulsation produces $\sim\pm 100$ $R_\odot$ variations in radius.

The structure of the circumstellar envelope appears to be dominated by a stationary 800 K layer at a distance of the order of 10 stellar radii from the photosphere. This layer in χ Cyg was rapidly built up late in 1975 and then steadily diminished over the next three cycles. During this interval, it may have provided a reservoir both for material outwardly accelerated by radiation pressure on condensing dust and for material seen draining back down to the photosphere. The resonance lines at optical wavelengths arise in the first of these regimes, and the low excitation lines in the second. The 800 K reservoir itself seems the only plausible location for SiO maser features to originate; if this is in fact the case, then one might expect a correlation between the behavior of the 800 K layer and SiO maser characteristics.

The extended time series of observations analyzed in this paper could not have been obtained without the dedicated support of W. A. Lenz and J. C. Golson at the Mayall 4 m telescope. For the visual light curve and times of maxima, we are indebted to J. A. Mattei and the AAVSO observers.

REFERENCES

Barnes, T. G. 1970, in *Proceedings of the Conference on Late Type Stars*, ed. W. Lockwood and H. M. Dyck (Kitt Peak Obs. Contrib. No. 554), p. 137.
_____. 1973, *Ap. J. Suppl.*, **25**, 369.
Bernat, A. P., Hall, D. N. B., Hinkle, K. H., and Ridgway, S. T. 1979, *Ap. J.* (*Letters*), **233**, L135.
Cahn, J. H., and Wyatt, S. P. 1978, *Ap. J.*, **221**, 163.
Campbell, L. 1955, *Studies of Long Period Variables* (Cambridge: AAVSO).
Christou, J., and Worden, S. P. 1980, *Ap. J.*, **85**, 302.
Dickinson, D. F., Reid, M. J., Morris, M., and Redman, R. 1978, *Ap. J.* (*Letters*), **220**, L113.
Dominy, J. F., Hinkle, K. H., and Lambert, D. L. 1982, in preparation.
Gehrz, R. D., and Woolf, N. J. 1971, *Ap. J.*, **165**, 285.
Gilman, R. C. 1969, *Ap. J.* (*Letters*), **155**, L185.
Hagen, W. 1978, *Ap. J. Suppl.*, **38**, 1.
Hall, D. N. B., Hinkle, K. H., and Ridgway, S. T. 1979, in *IAU Colloquium 46, Changing Trends in Variable Star Research*, ed. F. M. Bateson, J. Smak, I. H. Urch (Hamilton, New Zealand: University of Waikato Press), p. 264.
Hall, D. N. B., Hinkle, K. H., Ridgway, S. T., and Wojslaw, S. 1982, in preparation.
Hall, D. N. B., Ridgway, S. T., Bell, E. A., and Yarborough, J. M. 1979, *Proc. Soc. Photo-Opt. Instrum. Eng.*, **172**, 121.
Hill, S. J., and Willson, L. A. 1979, *Ap. J.*, **229**, 1029.
Hinkle, K. H. 1978, *Ap. J.*, **220**, 210.
Hinkle, K. H., Lambert, D. L., and Snell, R. L. 1976, *Ap. J.*, **210**, 684.
Johnson, H. R., Bernat, A. P., and Krupp, B. M. 1980, *Ap. J. Suppl.*, **42**, 501.
Joy, A. H. 1954, *Ap. J. Suppl.*, **1**, 39.
Kaifu, N., Buhl, D., and Snyder, L. E. 1975, *Ap. J.*, **195**, 359.
Labeyrie, A., Koechlin, L., Bonneau, D., Blazit, A., and Foy, R. 1977, *Ap. J.* (*Letters*), **218**, L75.
Maehara, H. 1971, *Pub. Astr. Soc. Japan*, **23**, 503.
Maillard, J. P. 1974, in *Highlights of Astronomy*, Vol. **3**, ed. G. Contopoulos (Dordrecht: Reidel), p. 269.
Mattei, J. A. 1979, personal communication.
Merrill, P. W. 1947, *Ap. J.*, **106**, 274.
_____. 1953, *Ap. J.*, **118**, 453.
_____. 1960, in *Stars and Stellar Systems*, Vol. **6**, *Stellar Atmospheres*, ed. J. L. Greenstein (Chicago: University of Chicago Press), p. 509.
Moran, J. M., Ball, J. A., Predmore, C. R., Lane, A. P., Huguenin, G. R., Reid, M. J., and Hansen, S. S. 1979, *Ap. J.* (*Letters*), **231**, L67.
Nather, R. E., and Wild, P. A. J. 1973, *Ap. J.*, **78**, 628.

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1982ApJ...252..697H

Norton, R. H., and Beer, R. 1976, *J. Opt. Soc. Am.*, **66**, 259.
Parsons, S. B. 1972, *Ap. J.*, **174**, 57.
Pierce, J. N., Willson, L. A., and Beavers, W. I. 1979, *Pub. A.S.P.*, **91**, 372.
Preston, G. W. 1964, *Ap. J.*, **136**, 866.
Reid, M. J., and Dickinson, D. F. 1976, *Ap. J.*, **209**, 505.
Ridgway, S. T., and Capps, R. W. 1974, *Rev. Sci. Instr.*, **45**, 676.
Scalo, J. M., and Ross, J. E. 1976, *Astr. Ap.*, **48**, 219.
Smak, J. I. 1966, *Ann. Rev. Astr. Ap.*, **4**, 19.
Snyder, L. E., and Buhl, D. 1975, *Ap. J.*, **197**, 329.
Thompson, R. I. 1977, *Ap. J.*, **212**, 754.
Wallerstein, G. 1959, *Ap. J.*, **130**, 560.
_____. 1975, *Ap. J. Suppl.*, **29**, 375.
_____. 1979, in *IAU Colloquium 46, Changing Trends in Variable Star Research*, ed. F. M. Bateson, J. Smak, I. H. Urch (Hamilton, New Zealand: University of Waikato Press), p. 264.
Willson, L. A. 1976, *Ap. J.*, **205**, 172.
Wood, P. R. 1979, *Ap. J.*, **227**, 220.

DONALD N. B. HALL, KENNETH H. HINKLE, and STEPHEN T. RIDGWAY: Kitt Peak National Observatory, PO Box 26732, Tucson, AZ 85726

© American Astronomical Society • Provided by the NASA Astrophysics Data System